Astronomy writing with LaTeX and AI/LLMs

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Abstract

As an astronomy educator, researcher, and outreach professional, I have been deeply engaged with astronomy literature for a long time. During my high school years, as I was learning the foundational concepts of astronomy, I came across countless notes written in LaTeX, each with unique styles of writing and presenting various concepts. Later, as I took on the role of team leader for Bangladesh's International Olympiad participants, I created numerous notes and problems using LaTeX to aid in preparation and question design. This eventually culminated in the writing of a textbook in Bengali. Another key experience came from participating in eight International Board Meetings (IBMs) of the IOAA, where leaders from 50 countries debated the nuances of writing, formatting, and symbolizing astronomical concepts. These discussions highlighted the need for a unified guide on writing astronomy with LaTeX. Recognizing this gap, I decided to create such a document—here it is. This document provides concise, practical rules and a template for writing self-contained, unambiguous, and well-formatted astronomy problems appropriate for national and international olympiads. It includes recommended LaTeX conventions, marking structure, AI/LLM usage checklist, and an example problem template that follows IOAA style conventions.

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1.1 Definitions

Definition (Self-containment)**.** A problem is *self-contained* if a solver with a standard high school or undergraduate physics/astronomy background (but possibly different subfield/olympiads i.e., IPhO/IJSO/IESO) can derive every step of the solution using only:

- The information explicitly provided in the problem statement, and
- Textbook-level knowledge and standard constants (e.g. Newton's gravitational constant G, speed of light c), when those constants are either given or their use is standard and explicitly permitted.

A self-contained prompt therefore requires that:

a) Every symbol appearing in the question and in subparts must be defined the first time it appears (units and numerical values included). For example: "Let $M_\star$ denote the stellar mass; take $M_\star = 1.2\,\mathrm{M}_\odot$."
b) Any observational or instrumental parameters (exposure time, filter central wavelength, telescope diameter, quantum efficiency, background sky brightness, etc.) needed to compute a numerical answer must be provided or the statement must explicitly instruct the solver to adopt a standard assumption and give its numerical value.
c) If a known physical constant is used, either give its numerical value in the problem statement (preferred) or explicitly state that the solver may use accepted CODATA values, table of constant [TOC] values and list which ones are allowed.
d) The problem explicitly states the approximation regime (e.g. "assume geometric optics", "neglect atmospheric refraction", "use non-relativistic mechanics"), or else must be exact within textbook methods.

Definition (**Unambiguity**)**.** A prompt is *unambiguous* if every expert reader interprets the task in the same way and that interpretation leads to a unique final answer (up to trivial algebraic rearrangements). Practically, ensure that:

a) There is no defensible alternative interpretation of a phrase or symbol that changes the final numerical result or the method required.
b) The statement forbids or explicitly allows common approximations when those approximations alter the result (for example: "Neglect limb darkening" or "Include limb darkening using model X with parameter values ...").
c) The final answer is fully specified: give required units and a directive on the form of the answer (e.g. numerical value, symbolic expression, order-of-magnitude estimate) and, for numerical parts, the number of significant figures required.
d) If vector or sign conventions matter (e.g. direction of positive angle, right-hand-rule for cross products, definition of position angle), state them explicitly.
e) Avoid trick-question phrasing that relies on linguistic ambiguity; olympiad problems should test physics and reasoning, not reading-comprehension traps.

1.2 Problem structure, marking, and expected time

Olympiad problems should be modular, progressively harder, and designed such that the marking scheme reflects the reasoning process rather than only the final answer. In IOAA-style examinations, the total time and marks are approximately matched (e.g. 300 marks in 300 minutes), implying that a well-prepared student is expected to spend roughly **1 minute per mark**. Thus, a 20-mark question is typically designed to be solvable in about 20 minutes by an IOAA-level student, while national-level students may require slightly longer (about 25–30 minutes).

Step-based marking philosophy. Each subpart should be decomposed into a sequence of conceptual and computational steps, with marks distributed across these steps rather than concentrated in the final answer. This ensures that partial understanding is rewarded and that a student who correctly models the physics but makes a minor algebraic or numerical mistake does not lose all marks. Such a structure reflects the core objective of IOAA theory problems: to evaluate **multi-step reasoning, physical modeling, and mathematical execution**, rather than isolated recall or single-step computation.[1]

In practice, IOAA problems are constructed so that each step corresponds to a distinct cognitive task:

- translating a physical situation into a mathematical model,
- combining ideas from different domains (e.g. geometry + astrophysics),
- performing controlled approximations,
- and executing calculations with appropriate precision.

Marks are therefore awarded at intermediate checkpoints, which often align with these reasoning transitions. This layered structure is essential because IOAA problems frequently require chaining multiple ideas across domains (e.g. celestial mechanics with radiative physics, or spherical geometry with observational constraints), and a student's performance must be evaluated at each stage of this chain.

A well-structured numerical subpart should implicitly follow the hierarchy:

1. **Physical setup (Model identification):** Identify the governing principle or equation (e.g. Newtonian gravity, Kepler's law, radiative transfer, hydrostatic equilibrium). *This step tests conceptual recognition.*
2. **Geometric or conceptual modeling:** Introduce a diagram, coordinate system, or physical interpretation where necessary. Many IOAA problems fall into geometric/spatial reasoning categories, where visualization is essential.[2]
3. **Algebraic development (Derivation):** Manipulate the equations into a usable or simplified form. This may include substitutions, approximations, or dimensional reasoning. *This step reflects the core mathematical reasoning component.*
4. **Substitution (Numerical execution):** Insert numerical values from the problem statement, ensuring consistent units and careful handling of constants. *This step evaluates computational discipline and precision.*

[1] IOAA theory problems are explicitly designed to probe multi-step derivations and integrated reasoning across physics and mathematics domains

[2] Approximately a significant fraction of IOAA problems involve geometric or coordinate-based reasoning, such as celestial sphere transformations and spherical trigonometry

5. **Final result (Validation):** Present the answer with correct units, appropriate order of magnitude, and required significant figures. Where possible, the result should be checked for physical plausibility. *This step distinguishes correct reasoning from accidental numerical agreement.*

Design implication. A well-designed IOAA subpart typically allocates **1–2 marks per step**, creating a natural progression of difficulty. Early steps are accessible and test foundational understanding, while later steps require deeper synthesis and careful execution. This progression aligns with observed IOAA difficulty distributions, where problems are intentionally structured to separate basic competence from advanced problem-solving ability.[3]

> **Key principle.**
>
> *Marks should reward the reasoning pathway — including modeling, structure, and intermediate logic — not only the final numerical answer.*

General Marking Rules used in IOAA (2023, 2024, 2024Jr)

Using incorrect physical concept (despite correct answers)	No points given
Giving correct answer without detailed calculation	Deduct 50% of the marks for that part
Minor mistakes in the calculations, e.g., wrong signs, symbols, substitutions	Deduct 20% of the marks for that part
Units missing from final answers	Deduct 0.5 pts
Too few or too many significant figures in the final answer	Deduct 0.5 pts
Error resulting from another error in an earlier part for which the student already lost marks, if the answer is physically reasonable	Full points (i.e., no deductions)
Error resulting from another error in an earlier part, where the student should have realised the answer was physically unreasonable.	Deduct 20% of the marks for that part

For example, if due to an error in an earlier part, the student calculates the mass of a star as 2×10^{30} kg instead of 2.5×10^{30} kg they will only lose marks for the earlier part. However, if, for the same reason, a student calculates the mass as 2×10^{25} kg, they should realize this is wrong (a few times the Earth's mass) and thus should lose some marks for this part as well.

[3] IOAA problem sets are constructed to span a wide range of difficulty levels, from straightforward applications to highly complex multi-step reasoning tasks .

Example: IOAA 2025, T3 (Gravitational Waves). The IOAA 2025 gravitational-wave problem provides a clear illustration of this philosophy:

https://ioaa2025.in/wp-content/uploads/2025/09/Theory-Solution.pdf

- **T3.1 (Orbital frequency derivation):**
  - Setup: Newtonian circular orbit relation $\omega^2 r = GM/(R+r)^2$.
  - Modeling: Use centre-of-mass relation $MR = mr$.
  - Simplification: Express ω in terms of total mass and separation.
  - Physical substitution: Replace separation using Schwarzschild radii.
  - Final answer: Numerical value of ω_{ini}.
- **T3.2 (Dimensional analysis):**
  - Setup: Write general scaling form.
  - Modeling: Assign dimensions to each quantity.
  - Algebra: Equate powers of M, L, and T.
  - Final result: Determine exponents (α, β, δ).
- **T3.3 (Time to merger):**
  - Setup: Differential equation for f_{GW}.
  - Modeling: Integration limits from initial to divergence.
  - Algebra: Perform integration and isolate t_{merge}.
  - Substitution: Insert M_{chirp} and ω_{ini}.
  - Final answer: Numerical estimate ($\sim 0.10\,\text{s}$).

Each of these parts distributes marks across multiple reasoning steps (typically 1–2 marks per step), rather than assigning all marks to the final expression. This ensures that the problem tests layered understanding: physical intuition, mathematical formulation, and computational execution.

Precision and answer format. For numerical subparts, the required precision must always be stated explicitly. For example:

Round your answer to 3 significant figures.

If intermediate rounding is allowed or if specific approximations (e.g. neglecting eccentricity or assuming circular motion) are required, these must be clearly stated in the problem.
In IOAA-style marking, numerical answers are not treated as a single exact value, but rather as belonging to a **range of acceptable answers**, depending on the level of precision and constants used by the student. This is formalized using a tolerance table.

Category	Accepted Range	Interpretation
Half credit (lower limit)	Broad lower bound	Correct method, but significant rounding or approximation errors
Full credit range	Narrow interval around true value	Correct method with consistent constants and precision
Half credit (upper limit)	Broad upper bound	Correct structure, but overestimation or accumulated numerical error

This structure reflects a key IOAA philosophy:

- Numerical answers depend on choices such as:
  - number of significant figures used for constants,
  - intermediate rounding,
  - approximations made during derivation.
- Therefore, a range is defined to:
  - reward physically correct reasoning even with minor numerical deviations,
  - penalize only when deviations indicate conceptual or structural mistakes.

For example, in IOAA 2025 (T3.1), the same physical solution yields:

- $138.084\,\mathrm{rad\,s^{-1}}$ using full-precision constants,
- $136.996\,\mathrm{rad\,s^{-1}}$ using 2 significant figures,
- $131.126\,\mathrm{rad\,s^{-1}}$ using 1 significant figure.

These values all arise from the *same correct derivation*, and thus are grouped into a grading range rather than treated as separate answers.

Key principle.

Marks should reward the route to the answer, not only the answer itself.

The tolerance-table system ensures that grading reflects understanding of the physics and mathematical structure, rather than sensitivity to minor numerical choices.

1.3 Using AI / LLMs responsibly

See More: *Large Language Models Achieve Gold Medal Performance at the International Olympiad on Astronomy & Astrophysics (IOAA)* – https://arxiv.org/abs/2510.05016

When using AI to draft problems and solutions, follow this checklist:

1. Specify the exact output format you want the model to produce (problem text, latex source, solution steps, numerical answers, mark allocation, hints). Include example tokens if necessary.
2. Run the problem through at least two independent models (e.g. ChatGPT, Claude, Gemini) and compare answers for consistency at the required significant-figure level.
3. If model outputs differ, inspect the question for hidden assumptions or ambiguities and correct them; rerun until all models (and a human reviewer) agree on the canonical solution.
4. Cross-check numerical constants and unit conversions in the solution; do not trust model-provided constants without verification.
5. For multi-part or long solutions, provide a compact, rigorous solution in LaTeX for graders to use; keep informal model prose separate from the official solution.

1.4 Creativity and originality

Good Olympiad problems usually combine standard physics with a novel twist. Recommendations:

- Use realistic but simplified astronomical contexts (observational setups, stellar models, orbital geometry) to give authenticity.
- Borrow pedagogical motifs from past IOAAs but change parameters, add a second constraint, or combine subfields (e.g., dynamics + radiative transfer) to make a fresh problem.
- Avoid copying published problem texts; if inspired by a past problem, explicitly note the inspiration for internal review.
- If inspired or taken any manner of data from a published paper, always quote the source in the question statement, the problem title, or the arXiv code should be given.

1.5 Template: problem and solution structure

Problem template (IOAA style). An IOAA-style problem should be written as a sequence of clearly defined, modular subtasks, each targeting a specific stage of reasoning. The structure should ensure that all necessary information is provided, all symbols are defined, and each subpart has a well-defined objective and marking scheme.
A standard problem format is:

(a) **Physical setup:** Introduce a concise but complete scenario with all required numerical values and definitions of symbols. The setup should be self-contained and unambiguous. [Marks]

(b) **Structured subtasks:** Each subpart should correspond to a distinct step in the reasoning chain (e.g. deriving a quantity, interpreting a result, or computing a numerical value). For numerical subtasks, always specify:

   - required units,
   - number of significant figures,
   - expected form of the answer (e.g. symbolic, numerical, ratio).

   Example:

*Compute the stellar luminosity L in watts. Round your answer to **2 significant figures**.*

[Marks]

(c) **Consistency instruction:** Conclude the problem with a global instruction to avoid ambiguity:

For numerical parts, round your answers to the number of significant figures specified in each subpart.

Design principle. Each subpart should map onto a specific reasoning step (modeling, derivation, computation, or interpretation), ensuring that marks are distributed across the full solution pathway rather than concentrated at the end.

Example (BDOAA-style)**.** The star Dschubba (δ Sco) has a parallax $p = 8\,\text{mas}$. Assuming it is a spherical blackbody with radius $R = 7.5\,R_\odot$ and surface temperature $T = 28{,}000\,\text{K}$, compute:

(a) Luminosity in W and in $L_\odot$ [3 s.f.] [2]
(b) Apparent magnitude [3 s.f.] [2]
(c) Absolute magnitude [3 s.f.] [2]
(d) Distance modulus [2 s.f.] [2]
(e) Peak wavelength $\lambda_{\max}$ [4 s.f.] [2]
(f) If the flux increases by Δf with $\Delta f \ll f$, determine the change in magnitude [4 s.f.] [2]

Solution outline (IOAA style). The solution should be presented as a clean, logically ordered derivation, following the same step-based philosophy used in marking.
Each subpart should be structured as:

1. **Model identification:** State the governing equation or physical principle.
2. **Interpretation/geometry:** Introduce any required diagram or physical reasoning.
3. **Derivation:** Perform algebraic simplification or transformation.
4. **Substitution:** Insert numerical values with consistent units.
5. **Final answer:** Report the result with correct units, order of magnitude, and required significant figures.

All assumptions and approximations (e.g. blackbody assumption, small-angle approximations, neglect of extinction) should be explicitly stated. Where relevant, intermediate expressions should be kept symbolic until the final substitution step to preserve clarity and reduce numerical errors.

[Philosophy: first state the modeling choice. Since no bandpass or photometric zero-point is given, interpret "magnitude" as *bolometric* magnitude.]
We use the standard constants

$$R_\odot = 6.957 \times 10^{8}\,\text{m}, \qquad L_\odot = 3.828 \times 10^{26}\,\text{W}, \qquad M_{\text{bol},\odot} = 4.74,$$

and

$$\sigma_{\text{SB}} = 5.670374 \times 10^{-8}\,\text{W}\,\text{m}^{-2}\,\text{K}^{-4}, \qquad b = 2.89777 \times 10^{-3}\,\text{m}\,\text{K}.$$

(a) **Luminosity.**

[Philosophy: identify the governing model first, then substitute numbers only at the end.]
For a spherical blackbody,

$$L = 4\pi R^2 \sigma_{\rm SB} T^4.$$

With

$$R = 7.5\,R_\odot = 7.5(6.957 \times 10^8)\,\mathrm{m}, \qquad T = 2.80 \times 10^4\,\mathrm{K},$$

we obtain

$$\begin{aligned} L &= 4\pi(7.5R_\odot)^2 \sigma_{\rm SB} T^4 \\ &\approx 1.19 \times 10^{31}\,\mathrm{W}. \end{aligned}$$

In solar units,

$$\begin{aligned} \frac{L}{L_\odot} &= \frac{1.19 \times 10^{31}}{3.828 \times 10^{26}} \\ &\approx 3.11 \times 10^4. \end{aligned}$$

$$\boxed{L \approx 1.19 \times 10^{31}\,\mathrm{W} \approx 3.11 \times 10^4\,L_\odot}$$

[Philosophy: report both dimensional and normalized answers, since the problem explicitly asks for both.]

(b) **Apparent magnitude.**
[Philosophy: use the luminosity-distance chain. The star's brightness is first converted to an absolute magnitude, then shifted by the distance modulus.] The parallax is

$$p = 8\,\mathrm{mas} = 8 \times 10^{-3}\,\mathrm{arcsec},$$

so the distance is

$$d = \frac{1}{p} = \frac{1}{0.008}\,\mathrm{pc} = 125\,\mathrm{pc}.$$

The bolometric absolute magnitude is

$$\begin{aligned} M_{\rm bol} &= M_{\rm bol,\odot} - 2.5\log_{10}\left(\frac{L}{L_\odot}\right) \\ &= 4.74 - 2.5\log_{10}(3.11 \times 10^4) \\ &\approx -6.49. \end{aligned}$$

The distance modulus is

$$\begin{aligned} \mu = m - M &= 5\log_{10}\left(\frac{d}{10\,\mathrm{pc}}\right) \\ &= 5\log_{10}(12.5) \\ &\approx 5.48. \end{aligned}$$

Hence the apparent magnitude is

$$\begin{aligned} m &= M_{\rm bol} + \mu \\ &\approx -6.49 + 5.48 \\ &\approx -1.01. \end{aligned}$$

$$\boxed{m \approx -1.01}$$

[Philosophy: keep the symbolic formula until the last line to minimize arithmetic loss and make the marking transparent.]

(c) **Absolute magnitude.**

[Philosophy: even if computed earlier, this subpart must independently show the full calculation to earn marks.]

The bolometric absolute magnitude is given by

$$M_{\text{bol}} = M_{\text{bol},\odot} - 2.5\log_{10}\left(\frac{L}{L_\odot}\right).$$

Substituting,

$$\begin{aligned} M_{\text{bol}} &= 4.74 - 2.5\log_{10}(3.11 \times 10^4) \\ &= 4.74 - 2.5(\log_{10} 3.11 + 4) \\ &= 4.74 - 2.5(0.493 + 4) \\ &= 4.74 - 2.5(4.493) \\ &= 4.74 - 11.23 \\ &\approx -6.49. \end{aligned}$$

$$\boxed{M_{\text{bol}} \approx -6.49}$$

[Philosophy: explicitly expanding the logarithm step avoids hidden arithmetic and makes marking granular.]

(d) **Distance modulus.**

[Philosophy: geometry (distance) must be derived independently from parallax before applying the magnitude relation.]

From parallax,

$$p = 8\,\text{mas} = 8 \times 10^{-3}\,\text{arcsec}, \quad d = \frac{1}{p} = \frac{1}{0.008} = 125\,\text{pc}.$$

The distance modulus is

$$\mu = 5\log_{10}\left(\frac{d}{10\,\text{pc}}\right).$$

Substituting,

$$\begin{aligned} \mu &= 5\log_{10}(12.5) \\ &= 5(1.097) \\ &\approx 5.48 \\ &\approx 5.5 \quad (2 \text{ s.f.}). \end{aligned}$$

$$\boxed{\mu \approx 5.5}$$

[Philosophy: rounding is applied only at the final step according to the required significant figures.]

(e) **Peak wavelength.**
[Philosophy: use the simplest local relation after the physical model is fixed. Wien's law is a one-step inference once blackbody emission is assumed.] By Wien's displacement law,

$$\lambda_{\max} = \frac{b}{T}.$$

Therefore,

$$\begin{aligned}\lambda_{\max} &= \frac{2.89777 \times 10^{-3}}{2.80 \times 10^{4}}\,\mathrm{m} \\ &\approx 1.035 \times 10^{-7}\,\mathrm{m} \\ &= 103.5\,\mathrm{nm}.\end{aligned}$$

$$\boxed{\lambda_{\max} \approx 103.5\,\mathrm{nm}}$$

(f) **Small flux increase and magnitude change.**
[Philosophy: linearize only after writing the exact expression. This rewards the correct expansion step and makes the approximation explicit.] Magnitude is defined by

$$m = -2.5 \log_{10} f + C.$$

If the flux changes from f to $f + \Delta f$ with $\Delta f \ll f$, then

$$\begin{aligned}\Delta m &= -2.5 \log_{10}\left(\frac{f + \Delta f}{f}\right) \\ &= -2.5 \log_{10}\left(1 + \frac{\Delta f}{f}\right) \\ &\approx -\frac{2.5}{\ln 10}\,\frac{\Delta f}{f} \\ &\approx -1.086\,\frac{\Delta f}{f}.\end{aligned}$$

$$\boxed{\Delta m \approx -1.086\,\frac{\Delta f}{f}}$$

[Philosophy: state the sign clearly. A brightness increase must produce a decrease in magnitude.]

Key principle.

A well-written solution mirrors the marking scheme: each step in the derivation corresponds to a markable unit of reasoning.

1.6 Submission and review checklist

Before submitting a problem to the national committee, verify:

- The problem is self-contained and unambiguous (run the checklist in 1.1).
- All symbols and constants are defined; units are consistent. (see table 1)
- AI outputs were cross-checked and any model disagreements resolved.
- Marking scheme is clear and divisible into small, testable steps. (run the checklist in 1.2).

2 LaTeX and notation conventions

Astronomy LaTeX typeset has its own rules and regulations. As we're doing academic work in Astronomy we should follow the conventions set by Astronomers. The basic typeset mistake and how to solve it can be found here. Some Astronomical writing Tips.

2.1 Useful Package

Here are some useful packages that you should install for Astronomical writings

```
\usepackage{mathabx} % Provides Astrological symbols
\usepackage[super]{nth} % Provides nth power superscript
\underpackage{mathtools} % Provides box environment inside align
\usepackage{pdfpages} % For inserting and addings pdfs
\usepackage{cancel} % For cencelling equations
\usepackage{mhchem} % checm equations
\usepackage{bm} % bold math
```

2.2 General Mistakes or Your working hard

Tip 1

The most common problem I've seen people hassle with is how to align/center your equations:
You can write like this

```
$$\cos a= \cos b \cos c+ \sin b \sin c \cdot \cos A$$
```

Which produces

$$\cos a = \cos b \cos c + \sin b \sin c \cdot \cos A$$

Double $$ sign enclosing will add extra spacing before and after the equation, but an easy way to do it is

```
\[\cos a= \cos b \cos c+ \sin b \sin c \cdot \cos A\]
```

$$\cos a = \cos b \cos c + \sin b \sin c \cdot \cos A$$

Tip 2

When we have multiple lines of equation instead of writing like this

```
$$\cos a= \cos b \cos c+ \sin b \sin c \cdot \cos A$$
$$\cos \zeta =\sin\delta \sin\phi +\cos\delta \cos\phi\cdot \cos H$$
```

Which produces

$$\cos a = \cos b \cos c + \sin b \sin c \cdot \cos A$$

$$\cos \zeta = \sin \delta \sin \phi + \cos \delta \cos \phi \cdot \cos H$$

We should simply use `\begin{gather*}` function

```
\begin{gather*}
   \cos a= \cos b \cos c+ \sin b \sin c \cdot \cos A\\
   \cos \zeta =\sin\delta \sin\phi +\cos\delta \cos\phi\cdot \cos H
\end{gather*}
```

Output:

$$\begin{gather*} \cos a = \cos b \cos c + \sin b \sin c \cdot \cos A \\ \cos \zeta = \sin\delta \sin\phi + \cos\delta \cos\phi \cdot \cos H \end{gather*}$$

Tip 3

Instead of using `\Rightarrow` in equation array use `\implies` or `\implied by`

```
\[\tan\theta = \frac{BT}{SB} \Rightarrow \phi= \theta= \arctan  \frac{BT}{SB}\]
```

Which produces

$$\tan\theta = \frac{BT}{SB} \Rightarrow \phi = \theta = \arctan \frac{BT}{SB}$$

```
\[\tan \theta=\frac{BT}{SB} \implies \phi= \theta= \arctan \frac{BT}{SB}\]
```

Output:

$$\tan\theta = \frac{BT}{SB} \implies \phi = \theta = \arctan \frac{BT}{SB}$$

Tip 4

Instead of using `<<` for **greater than** use `\ll` and for less than `\gg`

```
\[\frac{h}{R}<< 1\]
```

Which produces

$$\frac{h}{R} << 1$$

```
\[\frac{h}{R}\ll 1\]
```

Output:

$$\frac{h}{R} \ll 1$$

Tip 5

While writing step-by-step solutions/explaning equations you might need to box the final line or the answer part which is a norm. You can do this inside `align` or gather. Depending on the situation,

Using the gather, you can write

```
\begin{gather*}
i=\cos^{-1}\left(\frac{R_\odot}{0.5\; \rm AU}\right)\\
=\cos^{-1}\left(\frac{6.96\times 10^8}{0.5\times 1.5\times 10^{11}}\right)\\
\boxed{i=89.47^\circ}
\end{gather*}
```

Which produces

$$
\begin{gathered}
i=\cos^{-1}\left(\frac{R_\odot}{0.5\; \rm AU}\right)\\
=\cos^{-1}\left(\frac{6.96\times 10^8}{0.5\times 1.5\times 10^{11}}\right)\\
\boxed{i=89.47^\circ}
\end{gathered}
$$

These equations should be aligned for aesthetics reasons, but regular `userpackages` doesn't support $\boxed{x}$ and `align*` at the same time.

Solution is you install `mathtools` package, and inside align write `Aboxed`

```
\begin{align*}
i&=\cos^{-1}\left(\frac{R_\odot}{0.5\; \rm AU}\right)\\
&=\cos^{-1}\left(\frac{6.96\times 10^8}{0.5\times 1.5\times 10^{11}}\right)\\
\Aboxed{i&=89.47^\circ}
\end{align*}
```

$$
\begin{aligned}
i&=\cos^{-1}\left(\frac{R_\odot}{0.5\; \rm AU}\right)\\
&=\cos^{-1}\left(\frac{6.96\times 10^8}{0.5\times 1.5\times 10^{11}}\right)\\
\Aboxed{i&=89.47^\circ}
\end{aligned}
$$

2.3 Writing Units

- Use proper Astronomical/Astrological symbols as needed. For example: Temperature of a Star or Sun $\cancel{T_s}$, $\cancel{T_{sun}}$ instead $T_\star$, $T_\odot$ $\surd$
- One common mistake people make is leaving the Units in *italic* like kg. But all the units MUST be in normal text like **kg** but if you're using Astronomical units like $m_\odot$, $R_\oplus$ it's ok to keep them italic. Units which can be abbreviated should not be italic **parsecs** $\rightarrow$ $\cancel{pc}$ but **pc** $\surd$

Use this macros in your preamble to make life easy!

```
\def\ep{\epsilon}
\def\av#1{\langle#1\rangle}
\def\cm{{\rm \, cm}}
\def\mm{{\rm \, mm}}
\def\s{{\rm \, s}}
\def\erg{{\rm \, erg}}
\def\K{{\rm \, K}}
\def\g{{\rm \, g}}
\def\kg{{\rm \, kg}}
\def\km{{\rm \, km}}
\def\dyne{{\rm \, dyne}}
\def\Hz{{\, \rm Hz}}
\def\m{{\rm \, m}}
\def\asec{^{\prime\prime}}
```

- `\ep`: Defines `\ep` as ϵ, commonly used to represent a small quantity or emissivity.
- `\av{}`: Defines `\av{X}` to produce $\langle X\rangle$, representing an average or expectation value.
- `\Msun`: Defines `\Msun` as $M_\odot$, the solar mass, a standard astrophysical unit.
- `\Rsun`: Defines `\Rsun` as $R_\odot$, the solar radius.
- `\cm`: Defines `\cm` as cm, a unit of length in the CGS system.
- `\mm`: Defines `\mm` as mm, a smaller unit of length.
- `\s`: Defines `\s` as s, the unit of time (seconds).
- `\erg`: Defines `\erg` as erg, a unit of energy in the CGS system.
- `\K`: Defines `\K` as K, the unit of temperature in Kelvin.
- `\g`: Defines `\g` as g, the unit of mass in grams.
- `\kg`: Defines `\kg` as kg, the unit of mass in kilograms.
- `\km`: Defines `\km` as km, the unit of distance in kilometers.
- `\dyne`: Defines `\dyne` as dyne, a unit of force in the CGS system.
- `\Hz`: Defines `\Hz` as Hz, the unit of frequency.
- `\m`: Defines `\m` as m, the unit of distance in meters.
- `\asec`: Defines `\asec` as $''$, representing arcseconds, commonly used in astronomy.

Astrophysical Quantities

- The mass of the Sun is represented as `\Msun`, which outputs: $M_\odot$.
- The radius of a star can be written as `2\Rsun`, producing: $2\,R_\odot$.
- A distance of 100 cm is written as `100\cm`, resulting in: $100\,\mathrm{cm}$.

Physics and Astronomy

- A star's average temperature: `\av{T} = 6000\K` outputs: $\langle T \rangle = 6000\,\mathrm{K}$.
- The angular size of a celestial object: `0.3\asec` outputs: $0.3''$.
- Force acting on a body: `3\dyne` produces: 3 dyne.

```
\newcommand*{\jam}{\ensuremath{^{j}}}
\newcommand*{\h}{\ensuremath{^{h}}}
\newcommand*{\um}{\ensuremath{^{m}}}
\newcommand*{\us}{\ensuremath{^{s}}}
```

Usage:

- ^{j}: Superscript `j`, often used for indices or specific notations.
- ^{h}: Superscript `h`, used for hours.
- ^{m}: Superscript `um`, used for minutes.
- ^{s}: Superscript `us`, used for seconds.

Example:

- 10^{h} represents 10 hours.
- 15^{m} represents 15 minutes.
- 20^{s} represents 20 seconds.

Time Superscripts

These macros are for typesetting angles in degrees, arcminutes, and arcseconds.

```
\newcommand*{\de}{\ensuremath{^{\circ}}}
\newcommand*{\am}{'}
\newcommand*{\as}{''}
```

Usage:

- $^{\circ}$: Superscript degree symbol ($^{\circ}$).
- `'`: Prime symbol for arcminutes.
- `''`: Double prime symbol for arcseconds.

Example:

- 45° represents 45 degrees.
- $30'$ represents 30 arcminutes.
- $10''$ represents 10 arcseconds.

2.4 Writing Notation

Table 1: Greek Symbols used in Astronomy

α	Alpha	Right Ascension	ω	omega	Angular Speed, Argument of Periapsis
β	Beta	Ecliptic latitude, Speed as a fraction of light	ψ	psi	Wave function
γ	Gamma	Adiabatic Index, Stellar System (when used for velocity)	τ	Tau	Timescale, Half-life, Optical Depth
δ	delta	Declination, Small Change, Transit Depth	ϕ	Phi	Phase, Latitude, Azimuthal coordinate
ε	varepsilon	Emissivity	φ	varphi	Elongation
η	Eta	Efficiency, Mass fraction			
θ	theta	Arbitrary angle	Λ	Lambda	Cosmological Constant
κ	Kappa	Opacity	Σ	Sigma	Summation, Surface Brightness
λ	lambda	Wavelength, Ecliptic longitude, Length scale	Φ	Phi	Phase, Gravitational Potential, Photon Flux
μ	Mu	Distance Modulus (μ_d), Mean Molecular weight, Surface magnitude, Proper motion, Reduced Mass	Ω	Omega	Angular Speed, Density parameter, Solid angle, Longitude of ascending node
ν	Nu	Frequency, Neutrino	Π	Pi	Product, (Variable star) period
π	pi	3.14... Parallax angle [ϖ]	Ψ	Psi	Wave function, Phase angle
ρ	Rho	Mass density, Position angular distance	Δ	Delta	Change
σ	sigma	Stefan-Boltzmann constant, Cross section, Standard Deviation, Velocity Dispersion			

2.5 Renewcommands

```
% Define macros
\newcommand{\msun}{\ensuremath{M_\odot}}
\newcommand{\rsun}{\ensuremath{R_\odot}}
\newcommand{\lsun}{\ensuremath{L_\odot}}
\newcommand{\mearth}{\ensuremath{M_\oplus}}
\newcommand{\rearth}{\ensuremath{R_\oplus}}
\newcommand{\dearth}{\ensuremath{d_\oplus}}
\newcommand{\aearth}{\ensuremath{a_\oplus}}
\newcommand{\teff}{\ensuremath{T_\mathrm{eff}}}
\newcommand{\kB}{\ensuremath{k_\text{B}}}
\newcommand{\NA}{\ensuremath{N_\text{A}}}
\newcommand{\Fsup}{\ensuremath{F_\text{sup}}}
\newcommand{\dsup}{\ensuremath{d_\text{sup}}}
\newcommand{\thetasub}[1]{\ensuremath{\theta_{\text{#1}}}}
\newcommand{\mangle}[1]{\ensuremath{\measuredangle #1}}
\newcommand{\me}{\mathrm{e}}
\newcommand{\mi}{\mathrm{i}}
\newcommand{\diff}{\mathrm{d}}
\newcommand{\arc}[1]{{%
  \setbox9=\hbox{#1}%
  \ooalign{\resizebox{\wd9}{\height}{\texttoptiebar{\phantom{A}}}\cr#1}}}
```

Solar and Earth-related Quantities

- **\msun**: Represents the solar mass, $\mathrm{M}_\odot$. Example: $M_\odot = 1.989 \times 10^{30}\,\mathrm{kg}$.
- **\rsun**: Represents the solar radius, $R_\odot$. Example: $R_\odot = 6.96 \times 10^{8}\,\mathrm{m}$.
- **\mearth**: Represents the Earth's mass, $M_\oplus$. Example: $M_\oplus = 5.97 \times 10^{24}\,\mathrm{kg}$.
- **\rearth**: Represents the Earth's semi-major axis, $a_\oplus$. Example: $a_\oplus \approx 1\,\mathrm{AU}$.

Physical Constants and Quantities

- **\teff**: Represents the effective temperature, T_eff. Example: $T_\mathrm{eff} = 5778\,\mathrm{K}$.
- **\kB**: Boltzmann constant, k_B. Example: $k_\mathrm{B} = 1.38 \times 10^{-23}\,\mathrm{J/K}$.
- **\NA**: Avogadro number, N_A. Example: $N_\mathrm{A} = 6.022 \times 10^{23}\,\mathrm{mol}^{-1}$.

Miscellaneous Symbols

- **\thetasub**: Custom subscript for θ, e.g., θ_max produces θ_max.
- **\mangle**: Angle symbol using $\measuredangle$. Example: $\measuredangle ABC$ represents $\measuredangle ABC$.
- **\diff**: Upright differential operator, d. Example: $\int_a^b f(x)\,\mathrm{d}x$.

Mathematical Operations and Constants

- **\me**: Represents Euler's number, e. Example: e^x.
- **\mi**: Represents the imaginary unit, i. Example: $\mathrm{i}^2 = -1$.